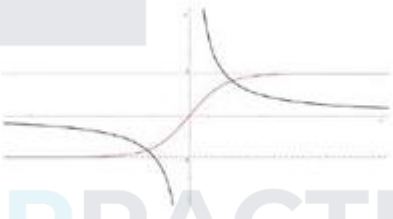


Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses quotient or product rule to obtain correct derivative	AO1.1b	B1	$y = \frac{x}{\cosh x} \Rightarrow \frac{dy}{dx} = \frac{\cosh x - x \sinh x}{\cosh^2 x}$
	Clearly sets 'their' $\frac{dy}{dx}$ numerator equal to 0	AO2.4	R1	Stationary point $\Rightarrow \frac{dy}{dx} = 0$ $\Rightarrow \frac{\cosh x - x \sinh x}{\cosh^2 x} = 0$
	Rearranges to complete a rigorous argument to show the required result. AG	AO2.1	R1	$\Rightarrow \cosh x - x \sinh x = 0$ $\Rightarrow \frac{\sinh x}{\cosh x} = \frac{1}{x}$ $\Rightarrow \tanh x = \frac{1}{x}$ AG
(b)	Sketches $\tanh x$ correctly including asymptotes	AO1.2	B1	
	Sketches $\frac{1}{x}$ correctly	AO1.2	B1	
	Deduces correct number of stationary points FT 'their' sketch in (b)(i)	AO2.2a	B1F	
(c)	Finds the second derivative	AO1.1a	M1	$\frac{d^2 y}{dx^2}$
	Obtains a correct expression for the second derivative	AO1.1b	A1	$= \frac{\cosh^2 x (\sinh x - x \cosh x - \sinh x)}{\cosh^4 x}$ $- \frac{2 \cosh x \sinh x (\cosh x - x \sinh x)}{\cosh^4 x}$
	Deduces that the second term is zero by using results from part (a)	AO2.2a	R1	second term is zero at stationary points

			$\frac{d^2y}{dx^2} = -\frac{x}{\cosh x} = -y$
Completes a rigorous argument to show the required result. AG Mark awarded if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$\Rightarrow \frac{d^2y}{dx^2} + y = 0$ AG
Total 10 marks			



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